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基于耳石微化学的长江下游刀鲚生境利用对策研究

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**Habitat Use Strategies of *Coilia nasus* from the Lower
Reaches of the Yangtze River based on Otolith
Microchemistry**

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Contents

CHAPTER 1 Literature review	1
1. Estuarine tapertail anchovy.....	1
2. The Yangtze River.....	2
3. Otolith research	3
3.1 Reconstruction of fish habitat use.....	3
3.2 Fishery management	4
3.3 Fishery conservation	5
3.4 Age determination	5
3.5 Life history	6
3.6 Recruitment evaluation	6
3.7 Stock identification	7
4. Justification of the study	8
CHAPTER 2 Habitat use and life history of <i>Coilia nasus</i> from the Chongming section of the Yangtze River.....	11
1. Introduction.....	11
2. Materials and methods	11
3. Results.....	14
4. Discussion	17
5. Conclusion	19
CHAPTER 3 Habitat use and life history of <i>Coilia nasus</i> from the Jiangjiang section of the Yangtze River	21
1. Introduction.....	21
2. Materials and methods	21
3. Result	22
4. Discussion	24
5. Conclusion	26
Chapter 4 Habitat use and life history of <i>Coilia nasus</i> from the Changshu section of the Yangtze River.....	27
1. Introduction.....	27
2. Materials and methods	27
3. Result	28
4. Discussion	30
5. Conclusion	31
GENERAL CONCLUSION	33

Innovations.....	35
References.....	37
APPENDIX.....	45

Fig. 2-3 Microscopic photographs of an otolith from <i>C. nasus</i> after sagittal section (transmitted light). 1: nucleus, 2: The first annual ring (1-year-old), 3: The second annual ring (2-year-old).....	16
Fig. 3-1 Fluctuation (a, b) and shift (A, B) of otolith Sr:Ca ratios along line transects from the core to the edge of <i>C. nasus</i> from Jingjiang section of the Yangtze River.	23
Fig. 3-2 Otolith Sr mapping analysis of <i>C. nasus</i> from Jingjiang section (sampling site). The values Sr concentrations corresponding 16 colors map from lowest blueish through greenish and yellowish to highest reddish.....	24
Fig. 4-1 Fluctuation (a, b) and shift (A, B) of otolith Sr:Ca concentration ratios along line transects from the core to edge of <i>C. nasus</i> from Changshu section of the Yangtze River.	29
Fig. 4-2 Otolith Sr mapping analysis of <i>C. nasus</i> samples from Changshu section. The values Sr concentrations corresponding 16 colors map from lowest bluish through greenish and yellowish to highest reddish.....	30

基于耳石微化学的长江下游刀鲚生境利用对策研究

摘要

长江下游是包含名贵鱼类刀鲚 (*Coilia nasus*) 在内的许多江海洄游型鱼类的主要分布区。更好地掌握相关水域刀鲚的洄游类型和生境利用对策知识, 对其未来资源的保护和管理十分必要, 也具有重要的意义。与此相适应, 本论文利用鱼类耳石微化学分析技术, 对采自长江下游(含河口) 江苏省靖江、常熟江段及上海市崇明江段刀鲚个体进行了生活史特征及生境利用对策的研究, 所取得的主要结果如下:

耳石微化学类型显示, 崇明江段刀鲚的生活史特征及生境利用对策非常复杂。其对淡水生境的利用程度出现了极为显著的差异。绝大多数个体的生活史过程中已具有了典型的河口半咸水或海水的生境履历, 但依其早期生活史对淡水生境的依存度, 又可分为短期和长期依存型两类。采样时, 其可能已经开始了溯河洄游过程。剩下的 2 尾个体尚无河口半咸水生境履历, 显示出采样时, 其从长江淡水水域刚移动到长江口, 尚未进入高盐度海水中进行下一步的生长和发育。崇明江段应该主要行使上述刀鲚资源重要洄游通道的作用。

靖江江段刀鲚个体均显示为典型的溯河洄游生活史类型, 但其对淡水生境的利用度也不尽相同。绝大多数个体在纯淡水生境中生活一段时间后才移动到河口, 但有 1 尾个体却较早地移动到近河口会受到海潮影响的淡水生境中生活。

常熟江段刀鲚个体亦均为典型的溯河洄游生活史类型。从其耳石对应生境的 Sr:Ca 比或 Sr 含量面分布反映出, 其早期生活史对淡水生境的依存同样可分为短期和长期依存型两类。但总体来看, 该江段刀鲚生活史特征及生境利用对策与上述崇明、靖江江段的有所不同。

关键词: 刀鲚; 锶; 钙; 耳石; 生活史; 生境利用; 耳石微化学; 长江下游

Habitat Use Strategies of *Coilia nasus* from the Lower Reaches of the Yangtze River based on Otolith Microchemistry

ABSTRACT

The lower reaches of the Yangtze River are the main distribution areas of anadromous fishes include high-value *Coilia nasus*. The knowledge of migration patterns and habitat use strategies of the *C. nasus* species in this water area is very necessary for conservation and management in the future. In this study, otolith microchemistry analysis was used to determine life history and habitat use of *Coilia nasus* of the Chongming section in Shanghai City and the Jingjiang and Changshu sections in Jiangsu Province in the lower reaches of the Yangtze River. The main results are as follows:

Life history and habitat use of *C. nasus* from Chongming section were very complex. All fish in this study revealed that they used freshwater habitats to different degrees in their life history. Most of fish would further move to brackish waters and sea water and then move back in freshwater habitat. Two of them seemed to just migrant near estuary regions but not yet enter to estuarine brackish water into brackish. This study revealed that the Chongming water in estuary of the Yangtze River would be important migratory routes for both *C. nasus* with or without anadromous life history.

The *C. nasus* samples collected from Jingjiang section of the Yangtze River anchovies were typical anadromous *C. nasus* ecotype, even they could use different strategies of freshwater habitats in their life history before they moved to brackish water habitats. Especially, one fish stayed in freshwater after that migrated near to the freshwater habitats that effect by tide or flooding marine water (high salinity).

The *C. nasus* collected from water around Changshu section are anadromous pattern of movement that utilized freshwater habitat in the early life then migrant to brackish water or seawater habitats of life history. The habitat use and life history characteristic are different with

the aforementioned fish from Jiangjiang section and Chongming section.

KEY WORDS: *Coilia nasus*; strontium; calcium; otolith; life history; habitat use; otolith microchemistry; lower reaches of the Yangtze River

ABBREVIATIONS

Ca: Calcium

CMCN: Chongming *Coilia nasus* (*C. nasus* collected from Chongming section)

CSCN: Changshu *Coilia nasus* (*C. nasus* collected from Changshu section)

C. nasus: *Coilia nasus*

EDTA: Ethylenediaminetetraacetate

EPMA: Electron probe microanalyzer

FFRC: Freshwater Fisheries Research Center

JJCN: Jingjiang *Coilia nasus* (*C. nasus* collected from Jingjiang section)

NAU: Nanjing Agriculture University

SD: Standard deviation

SPSS: Statistical Package for Social Scientists

Sr: Strontium

Sr:Ca: Strontium to calcium ratio

WFC: Wuxi Fisheries College

μm: Micrometer

Chapter 1

Literature review

1. Estuarine tapertail anchovy

The estuarine tapertail anchovy *Coilia nasus* (Temminck et Schlegel, 1846) mainly distributes in certain coastal, estuarine, and inland waters in China, Korea, and Japan (Liu et al., 2018) (Fig. 1-1). In China, *C. nasus* is a small sized anadromous, famous traditional delicacy, and highly commercial species and found in some large rivers, e.g., the Huanghe River (Yellow River), Changjiang River (Yangtze River), Qiantangjiang River, Oujiang River, and their estuaries, as well as river-connected lakes (like the Poyang Lake, Dongting Lake, Dongping Lake) (Jiang et al., 2014; Liu et al., 2018; Sokta et al., 2020). Especially, this anchovy is very important harvested fish in the mainstream, affiliated lakes and estuary of the Yangtze River (Yang et al., 2015) (Fig. 1-1). The fishing season of *C. nasus* in Yangtze River Estuary started earliest in February and latest in April, and the peak-fishing season usually continued from late March to middle April (Liu et al., 2012). Yield data of *C. nasus* in Yangtze River estuary in fishing season in 2010-2012 showed that the yield per boat in fishing season ranged from 65.9 to 875.4 kg with an average of 338.7 kg; the daily catch in fishing season ranged from 2.3 to 23.4 kg, averagely 9.4 kg; and the total licensed catch in fishing season ranged from 8.2 to 109.4 t, with an average of 42.3 t (Liu et al., 2012). The price of the anadromous *C. nasus* fish is much higher than that of the freshwater resident one (Jiang et al., 2017). It is probably the most expensive fish in the world with price as high as \$1000/kg and even \$9600 for a single extremely large individual with 45.3 cm total length of and 0.325 kg body weight (Yang et al., 2015). Anadromous *C. nasus* migrate from the sea to the rivers for spawning. The migratory season for this species is within a large window of time, from February to October (Sokta et al., 2020). The juveniles remain in freshwater habitats for 3–4 months until they acquire the ability to tolerate seawater; and then, follow the path of their parents and migrate to the sea (Xu et al., 2020). Three ecotypes of *C. nasus* have been found in the middle and lower reaches of the Yangtze River basin, i.e., freshwater resident, land-locked and anadromous ecotypes (Zhang et al., 2001; Liu et

al., 2014). The freshwater-resident and land-locked resources inhabit only freshwater, and spawns (Liu et al., 2014). In recently study, Chen (2016) indicated that not all long supermaxilla-type *C. nasus* in the Yangtze River migrate anadromously. However, this trait is traditionally believed as way used to distinguish the anadromous ecotype. Otolith strontium-to-calcium ratios of *C. nasus* have confirmed reflecting their habitat pattern throughout whole life history (Dou et al., 2012; D. Liu et al., 2014).



Fig. 1-1 Estuarine tapertail anchovy *Coilia nasus*.

2. The Yangtze River

The Yangtze River is the longest river in China and Asia. It stretches from Tibet to Shanghai (24°30'–35°45'N, 90°33'–122°25'E) where it finally flows into the East China Sea (Yang et al., 2007, 2009, 2011, Zhang et al., 2020). It is also the third longest river in the world, running 6300 km (3915 miles). Its basin, extending for some 3200 km (2000 miles) from west to east and for more than 1000 km (600 miles) from north to south (Muranov and Charles, 2019). The drainage basin covers a total area of $1808.5 \times 10^3 \text{ km}^2$ (Chen et al., 2001). Situated in the subtropical monsoon climate zone, the Yangtze floodplain is characterized by flooding during March–August coincident with maximum rainfall, and its hydrological regime is unique regionally (Chen and Chang, 1995; Ru and Liu, 2013). The Yangtze floodplain is among the largest floodplains in the world, with numerous shallow lakes (Shi and Qin, 2007; Ru and Liu, 2013). The river basin is usually divide into three parts, i.e. the upper reaches, the middle reaches and the lower reaches (Fu et al., 2003). The upper basin extends from the source of the river to Yichang (downstream of the Three

Gorges) (Chen et al., 2001). The middle reaches from Yichang to Hukou (Chen et al., 2001; Wang et al., 2020). The lower reaches extend from Jiujiang to the estuary (Chen et al., 2001). In the Yangtze River, historically 361–370 species and subspecies have been reported (Fu et al. 2003; Fu et al., 2004), however the latest research identified proximately 416 fish species and subspecies, 362 of which are strictly freshwater species (Ye et al., 2011). The number of endemic fish species is 178 (42.8%), out of which 65 are on the China Species Red List (Ye et al., 2011) (Zhang et al., 2020). The most species-rich phylogenetic orders in the river are Cypriniformes (280 species), Perciformes (50 species) and Siluriformes (40 species) (Zhang, 2020). Currently, among the 35 major freshwater aquaculture species in China, 26 species are distributed in the Yangtze River (Yang et al., 2007). The anchovy *C. nasus* is one of the most typical migration fish lived and spawned in this river. It has been believed as one national key protective species of China (Wang et al., 2020). To date, the resources of *C. nasus* has been adversely affected by human influences of pollution, fry by-catch, overfishing, and habitat degradation. (Yuan & Qin, 1984; Chen et al., 1999; Li, 1999; Liu et al., 2004; Ma et al., 2004). Furthermore, other human activities, including overfishing, water pollution, waterway construction and vessel navigation, river channelization, port construction, and sand/gravel extraction, have seriously affected the aquatic ecosystem of the Yangtze River (Yang et al., 2007, 2009, and 2011).

3. Otolith research

3.1 Reconstruction of fish habitat use

Otoliths ("ear-stones") are hard, calcium carbonate structures and found in the inner ear of bony fishes (Popper and Lu, 2000). There are three types of otoliths, all of which aid fish in balance and hearing. Both Sagittae and Asteriscus mainly functions in sound detection and hearing, and Lapillus mainly functions in detection of gravitational force and sound (Florida Fish and Wildlife Conservation Commission, 2021). The sagittae are always choose for element analysis as they are the largest type (Yang et al., 2011). Thorrold and Swearer (2009) showed that more recent studies concentrated on Ca, Sr, Mg, Mn, Ba and Pb, which are integrated from

the surrounding water and accumulate in fish body, as well as otoliths and other hard tissues. Otolith Sr:Ca ratio of concentrations has been especially used to reconstruct migrations of diadromous fish (Yang et al., 2011). In addition, otolith record the age and growth of a fish from the date of hatch to the time of death, as this material is precipitated from the endolymphatic fluid and is deposited continuously throughout the lifetime of the animal (Campana, 1999; Campana and Thorrold, 2001). Prior research has indicated that otoliths are suitably used as a record of the chemical environment experienced by an individual fish throughout its lifetime by their time-keeping properties, metabolic and temporal stability and the chronological signature in their concentric rings (Campana and Thorrold, 2001; Yang et al., 2011; Lisa and Campana, 2014).

Otolith microchemistry analysis reveals that the physicochemical characteristic of the water where the fish live was reflected by the otolith geographical signature. The advances have been reported to elucidate population connectivity (Gillanders et al. 2005), life history reconstruction (Elsdon et al, 2008), stock structure (Campana et al. 2000), migration dynamics (Kennedy et al. 2002), and natal tracking (Kennedy et al. 1997; Thorrold et al. 1998a, 1998b; Hobbs et al. 2005).

The freshwater, brackish water and seawater with different habitats used by fish (Yang et al., 2011). Many studies in previous time have showed that strontium (Sr) and calcium (Ca) microchemical signature in fish otolith could be used as a useful salinity scalar to estimate the migratory history, as otolith Sr and Sr/Ca ratio are robust markers of different salinity habitats of freshwater, brackish water and marine water (Limburg et al., 2011; Arai et al., 2002; Secor and Rooker, 2000; Campana, 1999; Tsukamoto et al., 2011; Arai et al., 2003; Dou et al., 2012; Jiang et al., 2014), and can be used on determination environmental history and natal origins of fish (Zeigler et al., 2010). In *Coilia species* their otolith Sr and Ca were analyzed using EPMA to estimate the habitats that they had lived in before they were captured and the microchemical data were then used to determine the migratory patterns of these fish species (Yang et al., 2006).

3.2 Fishery management

The total of capture fisheries production in the world was 90.9 million tons in 2016 (FAO, 2018). Otolith microchemistry can use to analysis life history of fish and assist law enforcement authorities for proof of illegal fishing activities such as illegal harvest or illegal stocking. That

strategic use of hard-part (including otolith) microchemistry methods can advance fish management and conservation across all stages of fish life history (Brenda, 2014), and use to identify the natal water body source of illegally introduced fish. Techniques of otolith chemistry along with other techniques also assist law enforcement authorities to determine fish species that exported or across regional boundaries by illegal with identify those illegal source of harvested or imported. Base on the otolith microchemistry analysis could be classifies hatchery and spawning ground of each resource population.

3.3 Fishery conservation

Fisheries and natural resource are faced with many challenges as fish stocks decline, over capitalized fishing fleets and illegal, unregulated and unreported (IUU) fishing undermines sustainability of fisheries worldwide. The effective conservation and management policy are the important majors to address the dwindling fisheries resources. Otolith microchemistry offers a powerful tool for inferring the environmental history and stock assignment of individual fishes and this technique to a wide range of fisheries conservation and management issues (Brenda, 2014). Because fish species use different environments in their life time, so it is important to understanding how the connection between the life cycles and habitats use of fish is importance for successfully conservation (Wilcove and Wikelski, 2008), and management. The otolith concentration could be used as a biological tag to discriminate among groups of fish that have spent their lives in different environments, moreover show the environmental conditions of spawning and nursery sites.

3.4 Age determination

The age information is important to understand the life history features and populations dynamics of fish (Beverton & Holt, 1957). One of the most serious mistakes made by fisheries biologists is the failure to validate age determination procedures (Beamish and Chilton, 1982). Age can be used to monitor spawning and hatching dates of fish to assessment environmental influences for growth condition and survival rate (Khan et al., 2014).

The fish age was determined by otolith, since these bony tissues often form yearly rings (annuli). The growth of otolith varies seasonally, which manifests optically as the banded macrostructure that is interpreted as annual increments to estimate fish age (Fowler, 2009). More

confirm from Fowler (2009) showed that the greatest application of otoliths to date has been in providing information on the age and growth of fish in years.

Otoliths of fish are hard, calcified internal structures that assist in orientation and sound perception (Popper et al. 2005; Green et al., 2009). As otoliths grow inside each fish's head they record an extraordinary amount of information about the life of that fish and the environments that it experiences (Begg et al., 2005). The width of each growth band that measured by daily otolith growth can be used to infer daily somatic growth (Jones, 1992). In fisheries, two important relationships have used as allometric relationship between size of fish and otolith and growth rate of individuals of fish or population show (increase per day of length or weight) (Jones, 1992). Otolith is one of the most important tools for understanding the age and growth of fish (Fowler, 2009). The otoliths continue to grow, at least in some dimensions, throughout the lives of the fish, as daily increments are added to the growing surface of the otolith (Campana & Neilson 1985; Morales, 2000).

3.5 Life history

Currently, the study of life histories, age of fish, ecological, population dynamic, and growth was very impressed. Otolith is a compound in the body of fish that has been recorded of fish daily growth, age and life history such as immigrate and change a habitat and presents a complete chronological record of the life of the fish (Campana, 1999; Bolle et al., 2004). it is possible to use chemical composition of otolith as natural tags because otolith are metabolically inert, continuous growth and incorporate trace elements as they accrete (Campana, 1999; Zheng et al., 2010). Analysis of this otolith microstructure has enabled us to examine many processes occurring during early life such as environmental influences on larval growth and survivorship, and the otolith microchemistry is used to analyze life history patterns that discriminate based on origin or nursery (Sponaugle, 2009; 2010; Arrhenius and Hansson, 1996).

3.6 Recruitment evaluation

The investigators can estimate individual's date of birth based on recent otolith daily ageing technique for recruitment calculations and survival-at-age (Jones, 1992). Likewise, in the stock system is used to replenish the fish population, significant challenges were found in determining the long-term survival and dispersion of fish after it has been released. Another important aspect

of predicting future fish abundance and distribution also relies on precise data related to fish population age structure, distribution, spawning habitats, and migratory patterns (Morris et al., 2003).

3.7 Stock identification

The fish-stock identification is a main basic for fisheries management (Cadrin et al., 2014). Element in otolith has been identified as a useful natural marker to identify fish stock structure (Lisa and Campana, 2014). The elements most likely to serve as environmentally influenced stock marker include Sr, Ba, Mn, Fe, and Pb (and perhaps Li, Mg, Cu, Ni, Se, and Hg), in which both ambient element: Ca concentrations and/or temperature produce significant effects on otolith composition (Fowler et al., 1995; Farrell and Campana, 1996; Geffen et al., 1998; Campana, 1999; Bath et al., 2000; Milton and Chenery, 2001; Lochet et al., 2010). The elemental isotope ratios that are most useful as natural tags are Sr, S, Pb, C, and O (Kennedy et al., 1997, 2000; Spencer et al., 2000; Thorrold et al., 2001; Weber et al., 2002; Kerr et al., 2007). Thorrold et al. (1997) showed that generally the combination of C and O isotopes used to indicate the severance in fish stocks caused by division during spawning. Fish stocks that inhabit different environments exhibit differences in the chemical composition of otoliths (Thresher, 1999; Secor et al., 2001; Campana, 2005; Elsdon et al., 2008). In particular, otolith elemental analysis may provide unique perspective on divergent migrations within stocks (Secor, 1999), and provide valuable information about stock structure and mixing patterns when the magnitude of environmental differences among areas is greater than the cumulative influence of any vital effects (Sturrock et al., 2012). In the previous studies, this approach was also used to determine the environment conditions during starting the first stage of life history (Campana et al., 1994; Gao et al., 2001). Thus, identification of fish stocks by otolith analysis would be necessary for some reasons of the distribution resources, protection and recognition of spawning and nurseries areas, and strategy for optimal harvesting (Begg et al., 1999).

4. Justification of the study

It is very important to understand the life history events of fish in order to effectively manage fisheries resources. Management of fisheries throughout the world relies heavily on understanding fish environmental history (Papetti et al., 2013; Wirgin et al., 2015). Migrant fish species is possible to meet the dynamics in different environments for seeking food and grown. Moreover, they change habitats area due to temperature condition or they need spawning ground. In the life history of fish spent variety habitats movement as like in freshwater and marine systems. It is note that the spawning or hatching area and life history habitats are necessary for fisheries conservation and management.

As aforementioned, there are many important reasons to choose otolith, because the otolith is a unique structure that records daily and annular events of a lifetime (Jones, 1992). Otolith microchemistry have been successful to study on natal origin of fish both sea and freshwater environments in recent years, and use to accurately classify migratory patterns and nursery areas of fish also. Chemical techniques use elemental signatures deposited permanently in otoliths can be used as natural tags for investigating diverse aspects of environmental history, including natal origins and movement (Carlson et al., 2016). Similarly, the content ratios of seven elements (Na, Mn, Fe, Ni, Cu, Sr, and Ba) to Ca in the nucleus indicated close relationship among *C. nasus* specimens collected from the Yangtze River Estuary, Poyang Lake, and Yellow Sea, suggesting a possible connectivity mechanism for anchovies in the three water bodies (Jiang et al., 2016).

The objective of this study was used the value Sr:Ca ratios on the otolith of each anchovy *C. nasus* from severer reprehensive locations in the estuary of the Yangtze River (Fig. 1-2) to reveal the life even and use the annual rings on the hart part to estimate the age of fishes. We hypothesize that anadromous *C. nasus* in the estuary of the Yangtze River constitute different degree of habitat use patterns and natal origins. The samples of *C. nasus* collected from the lower reaches of the Yangtze River at Jingjiang and Changshu sections in Jiangsu Province and Chongming section in Shanghai City) were analyzed using otolith microchemistry techniques to describe their habitat use and life history.

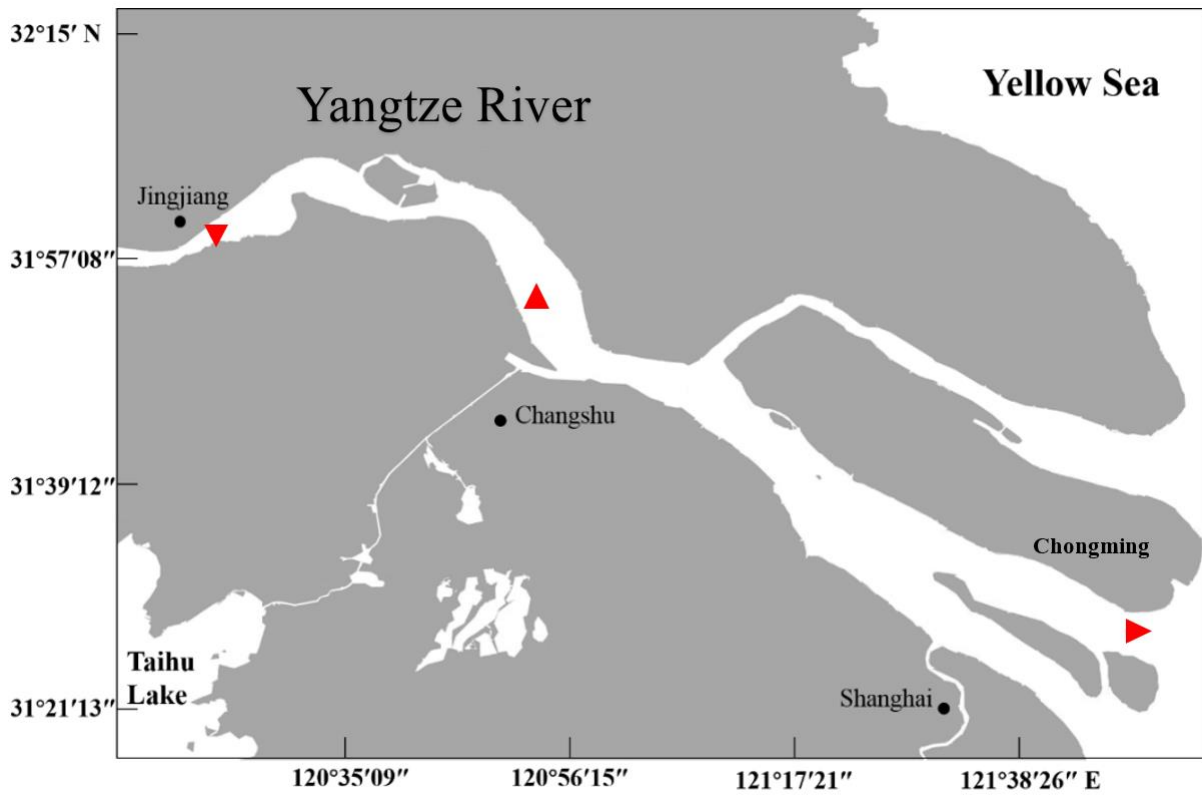


Fig. 1-2 showing of sampling site in three locations in the lower reaches of the Yangtze River. ▶: The site in the Chongming section, ▲: The site in the Changshu section, ▼: The site in the Jingjiang section

Chapter 2

Habitat use and life history of *Coilia nasus* from the Chongming section of the Yangtze River

1. Introduction

The estuarine tapertail anchovy *Coilia nasus* is a highly commercial species (Liu et al., 2018). The Chongming is the third largest island in China, and an alluvial island at the mouth of the Yangtze River. Chongming section lies on the route of anadromous migration of *C. nasus* (Yuan, 1987). From February onward, the adult *C. nasus* migrate from sea areas upstream to the lower and middle reaches of the Yangtze River and its conjoining lakes in the spawning season and return to the sea after spawning (Yuan et al., 1976). The previous studies have revealed that Sr and Ca signature in fish otolith could use as a useful scalar to estimate the migratory history (Jiang et al., 2014; Liu et al., 2018), as otolith Sr and Sr/Ca ratio are robust markers of different salinity habitats of freshwater, brackish water and seawater (Campana, 1999; Limburg et al., 2011; Secor and Rooker, 2000; Arai et al., 2002, 2003; Dou et al., 2012; Tsukamoto et al., 2011).

The aim of this chapter is to determine the life history characteristics of *C. nasus*, and establish its movement pattern and habitat use from the Chongming section in Yangtze River. All the results are hoped to understand well the ecotype composition of the *C. nasus* stocks and the possible habitat history and originated spawning ground locations of the fish in Chongming water area.

2. Materials and methods

The total of *C. nasus* of twenty-four samples (sample code of 19CMCN 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24) (Table 2-1) were analyzed in the present study with body weight 34.38 ± 11.47 g (mean $\pm$ SD, similarly hereinafter) and body length 238.41 ± 20.68 mm. All samples were collect in June 2019 from Chongming section in the river mouth of the Yangtze River at a location $121^{\circ}50'13''\text{E}$, $31^{\circ}24'31''\text{N}$. After biological

measurement, the anchovies were dissected to reveal the sex and gonad maturity reached at level (II to IV) (Table 2-1). The otoliths (left and right) of each fish was extracted and then cleaned well.

Water and ethanol 96% was used to clean otolith, keep it dried and stored it into the plastic tubes for more analysis (Fey and Jan, 2017). Digital microscope camera model QX800HD720P3D (China) with resolution 1280 x 720 has used to take picture of otolith and mounted it into a computer. The cleaned right and left otoliths of each anchovy were embedded by epoxy resin (EpoFix; Struers, Copenhagen, Denmark) and mounted on the glass slide. Using a cutting and grinding machine equipped with a diamond cup wheel (Discoplan-Ts; Struers, Copenhagen, Denmark) and further polished on a polishing wheel (LaboPol-35; Struers, Copenhagen, Denmark) to expose the core. After that, use ultrasonic bath and cleaned all otoliths with Milli-Q water and keep all samples in temperature drying box at 37°C overnight. Before examination in an EPMA, all otoliths were dried and carbon coated by high vacuum evaporator (JEE-420, JEOL Ltd Tokyo, Japan) (Jiang et al., 2012, Jiang et al., 2017; Khumbanyiwa et al., 2018). After EPMA analysis, carbon coating was cleaned from the otoliths and then they were etched by immersing them into 5% Ethylene Diamine Tetra Acetate (EDTA) to read the age of the fish by annulus rings.

Table 2-1 Sampling of *C. nasus* from the Chongming section in the Yangtze River Estuary

Sample Code	Total body length (mm)	Body weight (g)	Age estimated (years)	Sex and gonad development
19CMCN01	288	67.22	2 ⁺	♂ II
19CMCN02	257	42.82	2 ⁺	♂ II
19CMCN03	237	36.76	2	♀ III
19CMCN04	250	38.38	2 ⁺	♂ II
19CMCN05	225	26.93	1 ⁺	♂ II
19CMCN06	207	23.88	1 ⁺	♀ III
19CMCN07	221	23.71	1 ⁺	♀ III
19CMCN08	238	26.28	1 ⁺	♂ II
19CMCN09	243	30.97	1 ⁺	♀ III
19CMCN10	202	18.15	1 ⁺	♀ II
19CMCN11	234	30.03	2 ⁺	♂ II
19CMCN12	234	29.56	1 ⁺	♀ IV
19CMCN13	276	57.37	2	Undefined
19CMCN14	235	36.39	1 ⁺	♂ III
19CMCN15	222	27.52	1 ⁺	♂ II
19CMCN16	251	33.81	2 ⁺	♀ III
19CMCN17	228	29.1	2 ⁺	♀ III
19CMCN18	224	27.3	2	♀ III
19CMCN19	227	33.58	2	♂ III
19CMCN20	267	50.88	1 ⁺	Undefined
19CMCN21	242	31.3	1 ⁺	♀ III
19CMCN22	221	30.66	2	♀ III
19CMCN23	262	47.08	2	Undefined
19CMCN24	231	25.51	2 ⁺	♀ II

In this present study use the Regime Shift Detection Calculator, implements a new method based on a sequential algorithm developed by Rodionov (2004, 2005) for regime shifts in the mean level of fluctuations, is implemented in an Excel sheet to discriminate *C. nasus* life. Using the Mann-Whitney *U*-test tests the differences of otolith Sr:Ca ratios (Jiang et al., 2017).

By previous studies, otolith Sr:Ca ratios (i.e. Sr/Ca x 1000) and Sr content maps of *C. nasus* were found coinciding with different habitats of freshwater (low salinity, Sr:Ca ratio: ≤ 3 , X-ray intensity mapping color: blue, the same below), brackish (medium salinity, 3–7, green or yellow), and seawater habitats (high salinity, >7 , red), respectively (Yang et al., 2006; Jiang et al., 2014; Chen et al., 2017).

3. Results

Microchemical profiles of otolith of *C. nasus* collected from Chongming section in the Yangtze River indicated that a variable habitat uses three different color level patterns. According to the values of Sr:Ca ratio from the core to the edge along the life-history transect (Fig. 2-1) and the Sr color map pattern (Fig. 2-2) of each individual, *C. nasus* collected from Chongming section had divided in to three type. Type I observed in the otoliths of nine samples (19CMCN03, 19CMCN05, 19CMCN06, 19CMCN07, 19CMCN09, 19CMCN12, 19CMCN16, 19CMCN17, 19CMCN22). Type II observed in the otoliths thirteen individuals (19CMCN01, 19CMCN02, 19CMCN04, 19CMCN08, 19CMCN11, 19CMCN13, 19CMCN14, 19CMCN15, 19CMCN19, 19CMCN20, 19CMCN21, 19CMCN23, 19CMCN24). Type III had wider bluish regions freshwater of two fishes (19CMCN10 and 19CMCN18). The Sr:Ca ratios at the central otolith were as low as 2.56 ± 0.64 and 2.07 ± 0.61 for the Type I and Type II, respectively. In contrast, Sr:Ca ratios for whole otolith of Type III kept almost stable low values of 1.93 ± 0.67 .

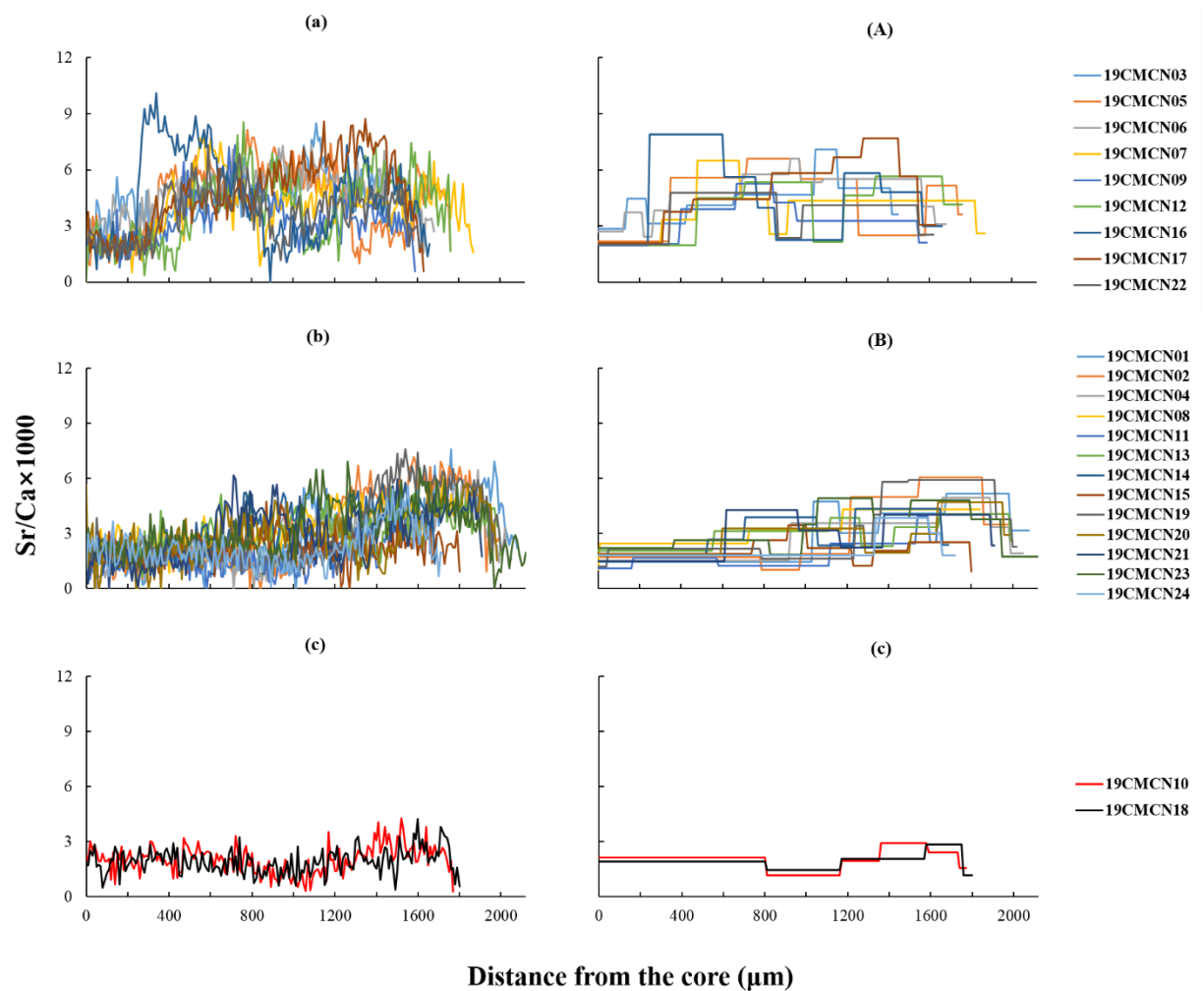


Fig. 2-1 Fluctuation (a, b, c) and shift (A, B, C) of otolith Sr:Ca concentration ratios along line transects from the core to the edge of *C. nasus* in Chongming section, Yangtze River.

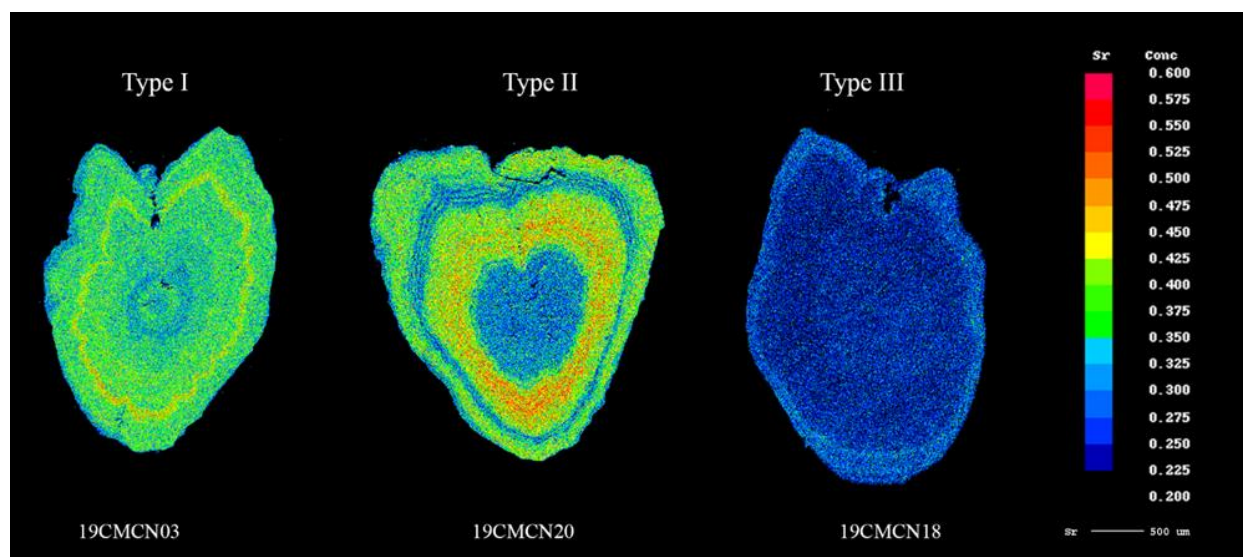


Fig. 2-2 Sr concentration maps of the otoliths of *C. nasus* samples from Chongming water area. The values Sr concentrations corresponding 16 colors map from lowest blueish through greenish and yellowish to highest reddish.

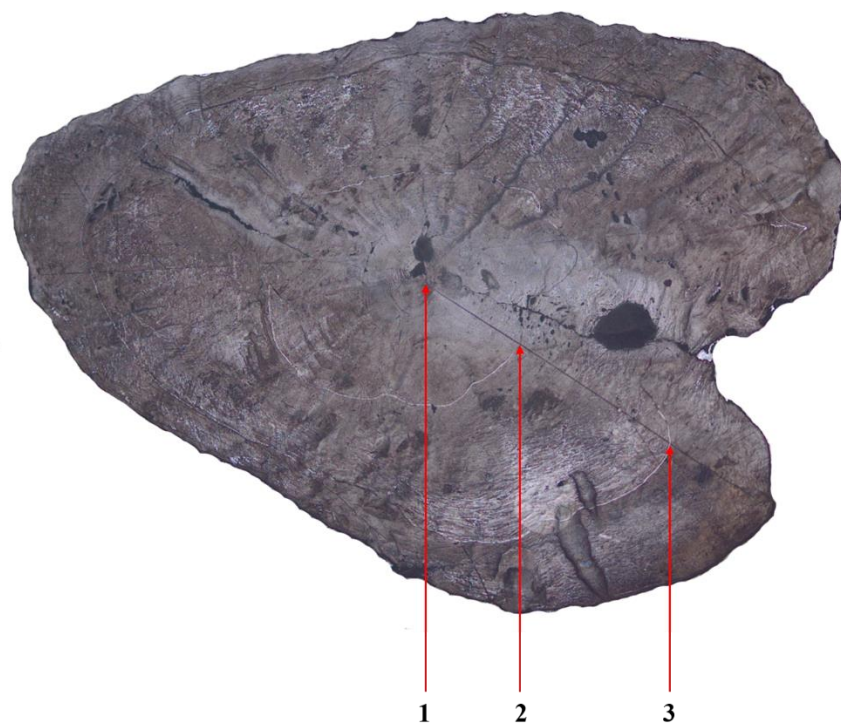


Fig. 2-3 Microscopic photographs of an otolith from *C. nasus* after sagittal section (transmitted light). 1: nucleus, 2: The first annual ring (1-year-old), 3: The second annual ring (2-year-old).

Viewing from Sr color map, Type I had shorter bluish form core regions to the point c. 70–480 μm (Fig. 2-1 (a), Fig. 2-2). Type II had wider bluish form the core to the point c. 560 – 1540 μm (Fig. 2-1 (b), Fig. 2-2). Type III had wider bluish regions freshwater of two fishes (19CMCN10 and 19CMCN18) (Fig. 2-1 (c), Fig. 2-2). Type III had bluish from core regions to the edge, showed that they spent almost whole lifetime in freshwater (Fig. 2-2).

4. Discussion

As aforementioned, the mouth and the main stream of the Yellow sea, Yangtze River and Taihu Lake already studied on freshwater resident and anadromous *C. nasus* (Chen et al., 2016; Jiang et al., 2012; Yang et al., 2006) have made reference of *C. nasus* of otolith Sr:Ca ratio and map markers of the salinity regimes of freshwater habitats (<3 , color: blue), brackish water (3–7, green or yellow), and seawater (>7 , red). Therefore, the low Sr:Ca ratios in the otolith showed that the fish spent life time in freshwater environment, while high Sr:Ca ratios showed fish living in seawater (Benson and Fitzsimons, 2002). Accordingly, the anchovies of Type I with low Sr:Ca ratios and shorter bluish core region, as well as Type II with also low Sr:Ca ratios and wider bluish regions at the central otolith indicated their riverine origin and initial freshwater residence. Type III kept low Sr:Ca ratios and low Sr bluish pattern throughout the otoliths, which mean their spent long period of lifetime in freshwater regions (Chen et al., 2016). Noteworthy, Type I and II also had higher Sr:Ca ratio areas with greenish, yellowish or reddish patterns at remaining regions neighboring the central regions of otoliths, indicating these individuals migrated into the high saline waters during corresponding periods of life history.

The patterns of life history of *C. nasus* from Chongming section indicated that much more variable and complex than previously thought. The wider bluish core regions hatched far from the estuary and shorter bluish regions hatched near the estuary (Khumbanyiwa et al., 2018). All nine fish of Type I with shorter bluish central core hatched near the estuary area (Fig. 2-2). All thirteen fish in Type II had longer bluish central core hatched far from the estuary (Fig. 2-2). It was observed that all samples of two types spent their early life in freshwater. After living in the freshwater habitat, both two types of fish changed Sr:Ca ratios around medium (3–7, blackish water) to low levels (≤ 3 , freshwater). Excepted three samples that from Type I, (19CMCN16)

changed to high levels (>7 , sea water) then moved to medium (3-7, brackish water) and to low levels (≤ 3 , freshwater) (keep moving back between medium to low throughout its life) (Fig. 2-1 (a)), and two remain (19CMCN03, and 19CMCN17) showed the environment changed between medium to high levels then move to medium levels again, lastly only 19CMCN17 moved between medium to low levels (Fig. 2-1 (a)). The edge of the otolith of 19CMCN03 with the value of Sr:Ca levels (3-7), which mean this fish just migrated to freshwater in short period then was soon caught and their offspring move to the estuaries, where they will remain until autumn, and then migrate to the ocean for growth and fattening. In previous study showed that, before moving into the brackish water or marine water *C. nasus* in Yangtze River hatched and spent in freshwater approximately 1 year. (Yang et al., 2006; Jiang et al., 2012). The anchovy drift in water into estuary and stay there until reached sexual maturity they migrated to upstream for spawning (Zhang, 2001; Li et al., 2007). In type III the Sr: Ca ratios were also consistently low (Fig. 2-1 (c)) along the otolith life history transect. Those species hatched far from the estuary (Fig. 2-2), and spent whole life history in freshwater regions. It was noted that the Type III had not experience in high salinity (>7). The two individuals of this type showed that near the end of life period they utilized habitat near freshwater slightly effected by tide or flooding from the seawater (relatively light bluish color around edge areas). It seems like fishes in this type live in the freshwater almost the full life and not yet migrated to estuary. Probably, the spawning grounds of these fish was very far from the estuary, and they spent much more time move to estuarine brackish water or marine water.

The estimation of age in this study indicated that, all anchovy age range between 1^+ and 2^+ years old. Noteworthyly, the length of time individuals in this study spend in freshwater is not directly proportional to the actual age of the individual. Jiang et al. (2014) give verified that elder male or female fish have a big otolith so bigger bluish central regions too. In the lower reaches of the Yangtze River, 89 % of anchovy spawners were found to be 2 years old fish (24–36 cm in body length) (Zhang et al., 2005). Thus, according to the age differences of each individual in this study the length of blue core regions and otoliths size had a little effected.

About 83% of the individuals in the present study showed the estimate of gonad maturity range from level II to III, the rest were estimated of level IV or undefined. The gonad maturity

in Type I reach of level III of nine individuals (37.5%) except two fishes, one sample (19CMCN05) reach to level II and other one remained (19CMCN12) even reached to level IV. All thirteen fishes (54.16%) of Type II reach at levels II, III and the rest was undefined, with in level II were seven individuals (29.16%) and levels III were three fishes (12.5%). The 19CMCN10 and 19CMCN18 of Type III reached a gonad maturity of level II and level III, respectively. This species reaches sexual maturity at 1-2 year-age (Ma et al., 2004), and breed once every year (Khumbanyiwa et al., 2018). The estimate of fish in this study on age from 1⁺ - 2⁺ indicated that most of these individuals already reached adult stage (Fig. 2-3). *C. nasus* makes an anadromous migration for spawning at approximately 2 years of age (Sokta et al., 2020). For those whom reached the gonad at lower level (II), they would be migrate upstream far from the captured site for spawning. Likewise, Ma et al. (2004), Yang et al. (2006) and Dou et al. (2012) showed in Yangtze River the annually adult *C. nasus* fish (anadromous fish) migrate from the estuaries to the middle and lower reaches of the river as well as affiliated freshwater lakes during the spring season for spawn.

5. Conclusion

Habitat use of *C. nasus* from Chongming section were very complex. All fish in this study revealed that they used freshwater habitats to different degrees in their life history. Most of fish would further move to brackish waters and sea water and then move back in freshwater habitat. A few of them seemed to just migrant near estuary regions but not yet enter to estuarine brackish water into brackish. This study revealed that the Chongming water in estuary of the Yangtze River would be important migratory routes for both *C. nasus* with or without anadromous life history.

Chapter 3

Habitat use and life history of *Coilia nasus* from the Jiangjiang section of the Yangtze River

1. Introduction

The Jingjiang section is located in the lower reaches of the Yangtze River and is a traditional fishery location for *Coilia nasus* (Wang, et al., 2019). The purpose of the chapter (Chapter 3) is to determine the habitat use strategies of *C. nasus* from the Jingjiang section in Yangtze River. It is hypothesized that all of *C. nasus* from Jingjiang section will have difference life history pattern with *C. nasus* from aforementioned Chongming section, even though this location is only about 180 km far from the latter.

2. Materials and methods

All individuals of *C. nasus* were collected from Jingjiang section of Yangtze River in May 2019 at a location 120°19'25"E, 31°58'10"N. There were ten fish (sample code: 19JJC01, 02, 03, 04, 05, 06, 07, 08, 09, and 19JJC10) in the present study with body weight of 117.65 ± 42.79 g and total body length of 327.8 ± 42.4 mm. The biological parameters of each *C. nasus* present in Table 3-1.

All the methodology for otolith treatment and determination, as well as for data analysis are as the same as those mentioned in 2.2 of chapter 2.

Table 3-1 Sampling of *C. nasus* from Jingjiang section in lower reaches of the Yangtze River

Sample code	Total body length (mm)	Body weight (g)	Age estimated (years)	Sex and gonad development
2019JJC01	364	154.94	2 ⁺	♂ IV
2019JJC02	345	122.11	2 ⁺	♀ III
2019JJC03	311	91.41	2 ⁺	♂ III
2019JJC04	290	75.87	2 ⁺	♂ II
2019JJC05	252	49.12	2	♀ III
2019JJC06	303	105.58	2 ⁺	♂ III
2019JJC07	301	90.04	2 ⁺	♀ IV
2019JJC08	384	188.01	2 ⁺	♂ IV
2019JJC09	366	146.44	2 ⁺	♂ III
2019JJC10	362	153.05	2 ⁺	♂ IV

3. Result

Ten fish of *C. nasus* collected from the Jingjiang section in the lower reaches of the Yangtze River had body length range from 252-384 mm and the body weight range from 75.87-188.01 g. The gonad maturities of them being reached level II to IV that most of them were in level III and IV (Table 3-1).

The microchemical analysis of otoliths indicated that the Sr:Ca ratios were complex variability and Sr maps. The results of *C. nasus* sampled showed similar habitat patterns in the early life history. The values Sr:Ca ratios of *C. nasus* divided the fish in to 2 types, the first type (Type I) had wider bluish central core (Fig. 3-2). Thereafter, the map profiles changed from bluish then to greenish and yellowish which shown the characteristic of anadromous fish. In Type I, nine fish (19JJC01, -02, -03, -04, -06, -07, -08, -09, and 19JJC10) presented bluish core regions from the core to point c. 520-1190 μm with low Sr:Ca ratios (< 3), which suggested a period of their life history in freshwater. Moreover, the longest radius of bluish central core

could be 1190 μm with low Sr:Ca ratio of 1.27 ± 0.59 in otoliths from Type I fish, suggesting that the fish hatched far from the estuary area and stay in freshwater habitat for long period of the early stage of life history. The Type II had interchanged bluish and greenish bands from the central core (Fig. 3-2). Fish in Type II was one individual (19JJC05). The history pattern hatched obviously farer from the estuary area than Type I, and stayed in freshwater habitat long time since was born then migrated to other areas.

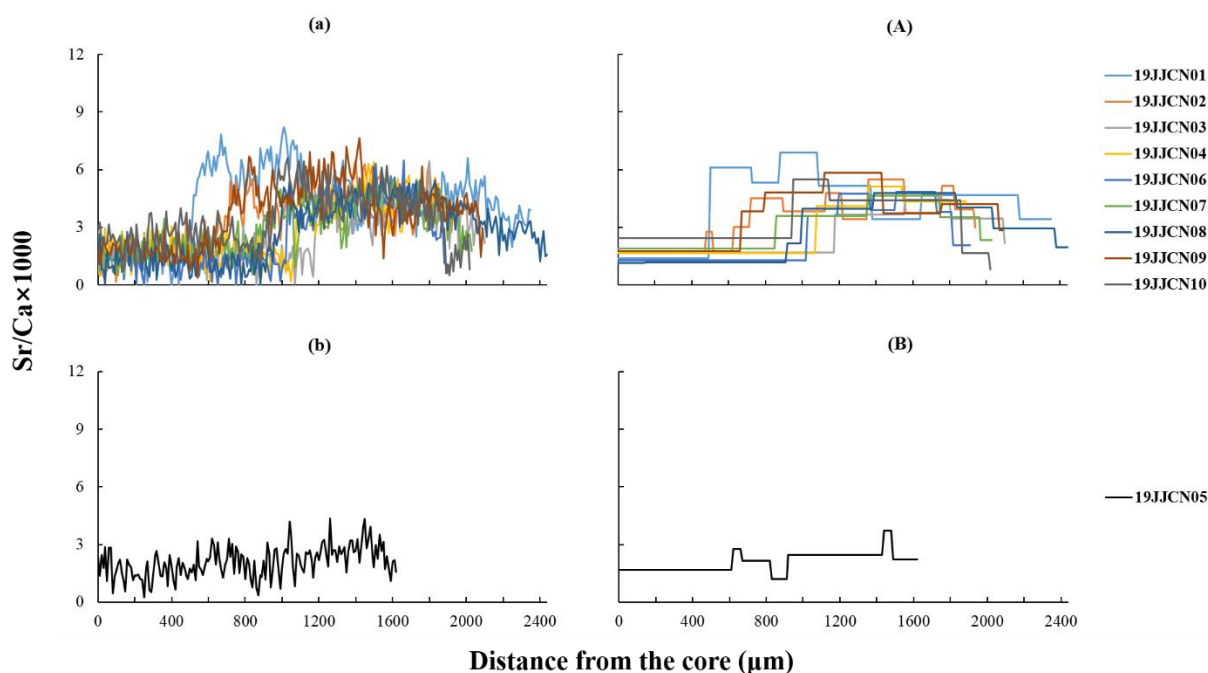


Fig. 3-1 Fluctuation (a, b) and shift (A, B) of otolith Sr:Ca ratios along line transects from the core to the edge of *C. nasus* from Jingjiang section of the Yangtze River.

The X-ray intensity maps showed that all otoliths of Jingjiang section of Yangtze had a similar bluish color (low Sr) in their central regions, which was consistent with the above results of the life history transect analyses. With 19JJC01, 19JJC02, 19JJC03, 19JJC04, 19JJC06, 19JJC07, 19JJC08, 19JJC09 and 19JJC10 were longer bluish core regions through Sr:Ca ratios low (< 3) (Fig. 3-2). The remaining part of samples had one individual (19JJC05) was complicated bluish-greenish regions (Fig. 3-2).

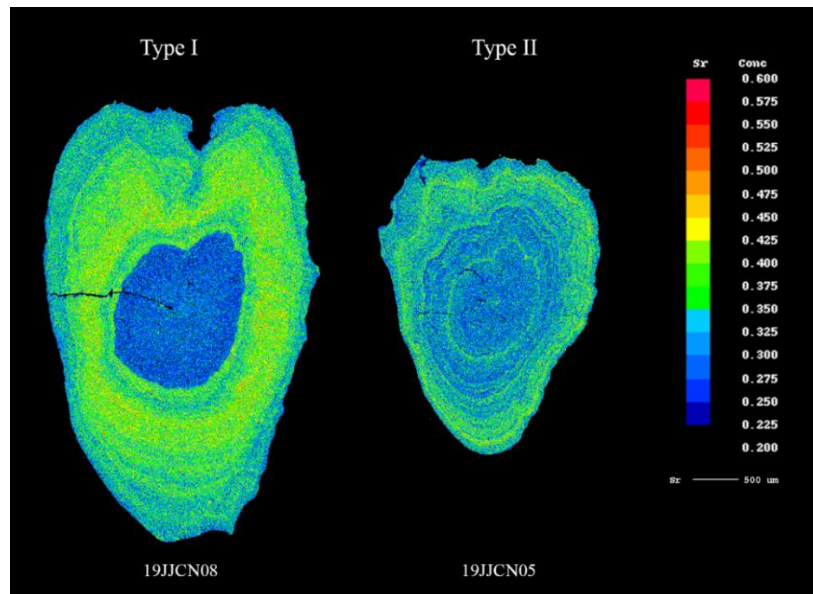


Fig. 3-2 Otolith Sr mapping analysis of *C. nasus* from Jingjiang section (sampling site). The values Sr concentrations corresponding 16 colors map from lowest blueish through greenish and yellowish to highest reddish.

4. Discussion

The study used Sr to Ca ratio to infer habitat shifts across salinity gradients (Dominique, 2019). Strontium is particularly useful for reconstructing environmental history of fishes as it replaces Ca in the otolith matrix according to its availability in the fish habitat (Pracheil et al., 2014; Secor and Rooker, 2000; Campana, 2005). The Sr:Ca ratios value could be showed the central core of the otolith regions of *C. nasus* species to understand the origin of fish and the early stage of freshwater living. Moreover, the otolith Sr:Ca ratio of concentrations has been especially used to describe diadromous migrations (Yang et al., 2011). This present study of *C. nasus* at Jingjiang section categorized into two types according to the size of the central bluish (short and long bluish core of the otolith area). The Type I had nine samples with wider bluish band of central core regions low Sr:Ca ratios. Thereafter, those group's fish alternated from low to medium Sr:Ca ratios values. The Type II had a complicated bluish-greenish bands throughout the whole otolith regions. Both of two types were spent early life in freshwater as like those studies of Chen et al. (2016), Jiang et al. (2012) and Yang et al. (2006). Hence, Type I was not hatched near the estuary and Type II was hatched from different spawning site with that of Type I.

The individuals in Type I was experienced in freshwater then the color patterns alternated among greenish or yellowish bands, corresponding to estuarine habitat histories (Liu et al., 2018). Thereafter, they migrated to freshwater again for spawning. It was noted that none of the samples had experienced in marine water with Sr:Ca ratio >7 (high salinity). It was suggesting that the fish hatched far the estuary area and stay long time in freshwater habitat of their life history since they was born. It was noted that, all nine individuals had similar life history living in freshwater and blackish water habitats. About ninety percent of the gonad maturity are relatively high from level III and IV.

One individual in Type II showed the interchanged color between bluish and greenish bands from core regions was $1440\text{ }\mu\text{m}$ with low Sr:Ca $\times 1000$ of 2.02 ± 0.78 in the otolith. Thus, value Sr:Ca ratios (low <3) indicated that fish spent long time of life history at first stage in freshwater and then changed to brackish water (3-7). The 19JJCNO5 seem like living in freshwater that effected by tide or flooding by the seawater, perhaps that place have enough foods for feeding and growing. After spent short period in brackish water an individual migrated back in freshwater habitat. The value of F_c was 0.88 higher than fishes in type I. It was estimated that fish hatched farer estuary than type I and migrated to other habitat stay long time in area effected by high salinity (seawater). The gonad maturity of 19JJCNO5 was level III.

The samples of this study were of age 2 to 2+ years old have not much effect of length of the bluish central core regions. The total length of samples in this study were (252-384 mm), Thus quite similar with Zhang et al. (2005) that about 89 percent (around 24-36 cm in body length) of anchovy spawners were found recently to be 2 years old fish in the Yangtze River, lower reaches part. About 10 percent the gonad maturities reached to level II (19JJCNO4), 50 percent of (19JJCNO2, -3, -5, -6 and 19JJCNO9) range at level III and 40 percent (19JJCNO1, -07, -8, and 19JJCNO10) at level IV. During a long spawning season ranging from May to August, adults ascend to the river from the sea for spawning (Whitehead et al., 1988; Liu et al. 2018). Thus, according to the estimated age and the gonad level include caught date in the study revealed that *C. nasus* from Jingjiang was returned to freshwater habitat and made anadromously migration to their spawning sites.

5. Conclusion

In this study the results indicated, all of the *C. nasus* samples collected from Jingjiang section in the lower reaches of the Yangtze River anchovies were typical anadromous *C. nasus* ecotype, even they used freshwater environment in vary degrees in life history events, after that fishes appear their migrated to brackish water. Some fishes stay in freshwater after that migrated near to the freshwater habitats that effect by tide or flooding marine water (high salinity). This result revealed that life history and habitat use of *C. nasus* in this section of the Yangtze River seemed to be different from the fish sampled from Chongming area in estuary of the river.

Chapter 4

Habitat use and life history of *Coilia nasus* from the Changshu section of the Yangtze River

1. Introduction

The Changshu section is located in the lower reaches of the Yangtze River between Jingjiang section and Chongming section. The Changshu section is about 37.5 km long and one of the main distributional areas and fishing grounds of *Coilia nasus*, being locations about 95 km away from the mouth of the river (Chen et al., 2016). The anchovies collected from the above area were studied by using EPMA to analysis of otolith Sr and Ca signatures, which can be applied as a useful scalar to estimate migratory history of fish (Brown & Severin, 2009; Limburg et al., 2011; Yang, Jiang, & Liu, 2011). The aim of this study is to understand habitat use strategies of *C. nasus* captured from Changshu section.

2. Materials and methods

Five samples of *C. nasus* (19CSCN01, 02, 03, 04, and 19CSCN05) collected from Changshu section with body weight of 76.92 ± 8.46 g and total body length of 315.4 ± 7.5 mm (Table 4-1). All fish were captured from the section around a location $120^{\circ}51'8''\text{E}$, $31^{\circ}52'31''\text{N}$ in June 2019. The biological parameter of each individual were recorded and measured, e.g., total body length, body weight, sex, gonad and maturity. All the methodology and data for otolith microchemistry analysis used in this chapter are the same in the above chapter (Chapter 2).

Table 4-1 Sampling of *C. nasus* from Changshu section in Yangtze River.

Sample code	Total body length (mm)	Body weight (g)	Age estimated (years)	Sex and gonad development
19CSCN01	319	74.4	2 ⁺	♀ II
19CSCN02	316	84.9	2 ⁺	♀ III
19CSCN03	303	86.7	2 ⁺	♀ III
19CSCN04	316	70.9	2 ⁺	♂ II
19CSCN05	323	67.7	2 ⁺	♂ III

3. Result

The *C. nasus* individuals caught from the Changshu section in the lower reaches of the Yangtze River was analysis the elements Sr and Ca base on the value of Sr:Ca ratio by otolith microchemistry analysis. The five fishes were typical anadromous individuals which hatched in freshwater habitat then moved to brackish water and seawater habitat for feeding. These fish could even be divided into two types based on different patterns of habitat use.

In Type I had shorter bluish central core with two individuals (19CSCN01 and 19CSCN02), from the core to point c. 70-250 μm (Fig. 4-1 (a)). The low Sr:Ca ratios lower than three (>3), mean that these fishes living in freshwater environment for a short time. After spent not much time in low salinity then they move to medium salinity where they develop growing and feeding with the referent of medium salinity (3-7), mean $\pm$ SD (6.51 ± 0.84) (19CSCN02), thereafter individuals keep alternate change from medium to low salinity, and the end of life fishes move back to low again then was soon captured. The 19CSCN02 show higher Sr:Ca ratio of 8.05 ± 0.58 , which mean had an experience in seawater environment.

Three fishes in Type II (19CSCN03, 04 and 19CSCN05) had longer bluish core regions from the core to point c. 610-970 μm with low Sr:Ca ratios of 1.23 ± 0.57) (19CSCN03) (Fig. 4-1 (b)), showed that individuals spent long time in freshwater than Type I. Thereafter, the Sr map patterns changed from bluish to greenish and yellowish (Fig. 4-2). All fishes in this type indicated the value of Sr:Ca ratios were low (<3) at the edge of the otolith of each sample,

excepted one fish (19CSCN05) had medium value Sr:Ca ratios of 4.41 ± 1.00 , meaning fish just move back in freshwater in very short time then soon captured.

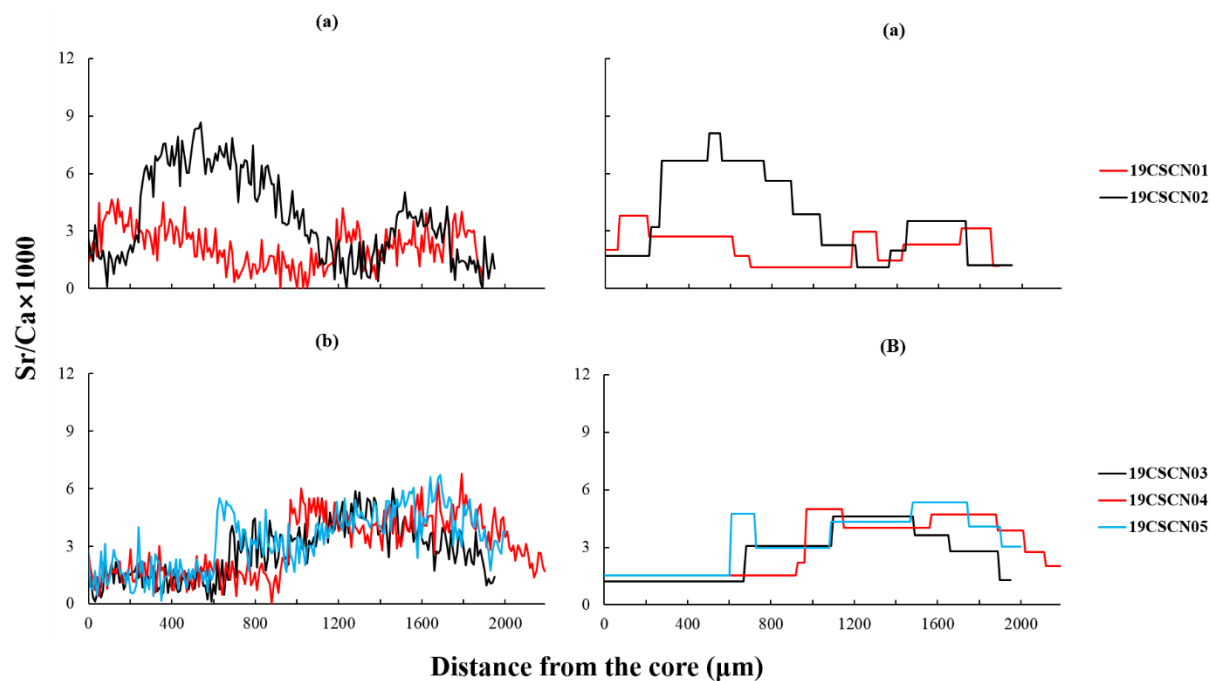


Fig. 4-1 Fluctuation (a, b) and shift (A, B) of otolith Sr:Ca concentration ratios along line transects from the core to edge of *C. nasus* from Changshu section of the Yangtze River.

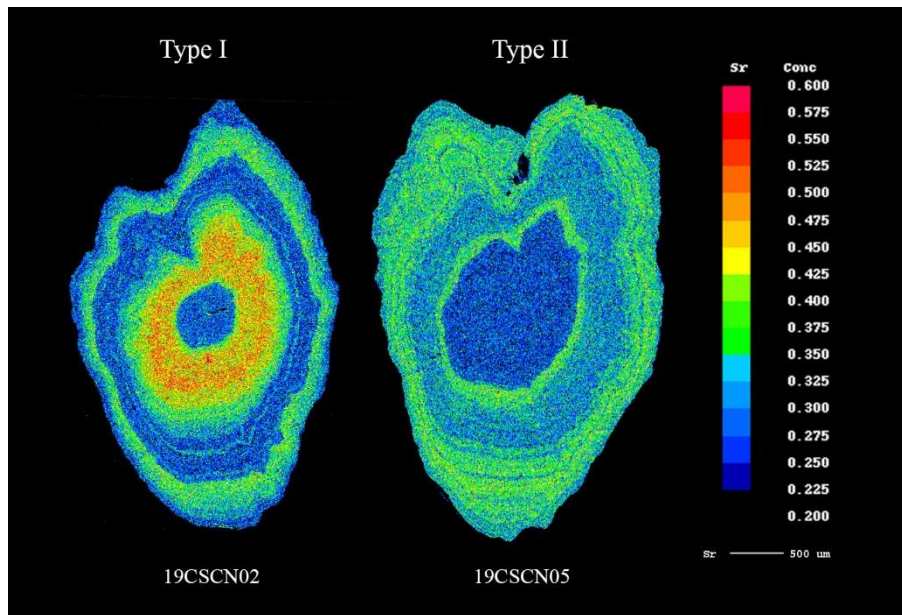


Fig. 4-2 Otolith Sr mapping analysis of *C. nasus* samples from Changshu section. The values Sr concentrations corresponding 16 colors map from lowest bluish through greenish and yellowish to highest reddish.

4. Discussion

Otolith of *C. nasus* of Changshu section in the lower reaches of the Yangtze River had bluish central core as like anchovy from Jingjiang section and Chongming section, showed that all of them originated from freshwater habitat since they were born. Most of *C. nasus* had spent long periods in freshwater environment before they migrated to other environments. As aforementioned, the salinity in water (freshwater, brackish water and seawater) were used to determine water habitats for fish (Yang et al., 2011). The freshwater habitat can be decided as $Sr:Ca \leq 3$ with Sr blue color mapping for low salinity, the brackish habitat can be decided as $Sr:Ca 3-7$ with Sr green or yellow color mapping for medium salinity, and sea water can be decided as $Sr:Ca > 7$ with Sr red color mapping for high salinity (Khumbanyiwa et al., 2018). By the way, this present study was divided into two types of *C. nasus* of Changshu section. The Type I had two samples were 19CSCN01 and 19CSCN02, the low $Sr:Ca$ ratio of 1.80 ± 0.66 (19CSCN02) had shorter bluish core regions from core to point c. 70-250 μm. Type I indicated that two fishes spent a short time in freshwater environment before they move to brackish water

and marine water. According to Khumbanyiwa et al. (2018), confirmed wider bluish core regions hatched far from the estuary and shorter bluish regions hatched near the estuary, thus two samples in Type I hatched near the estuary and after spent short periods in freshwater then they migrated to brackish water. The 19CSCN02 showed high Sr:Ca ratios of 8.05 ± 0.58 , which mean this fish was move to seawater in life history for growing and feeding.

The Type II were three fishes as 19CSCN03, -04 and 19CSCN05 and had wider bluish central core regions with low Sr:Ca ratios of 1.23 ± 0.57 (19CSCN03). Moreover, they hatched far from the estuary. All of them utilized freshwater environment for long time before they entering brackish water. The values of Sr:Ca ratio of the otolith showed from low salinity to medium salinity. Noted that three *C. nasus* in Type II had no experience in seawater. Individuals in the second type indicated the value of Sr:Ca ratios were low (<3) at the edge of the otolith of each sample. Excepted one fish (19CSCN05) showed medium value of Sr:Ca ratios of 4.41 ± 1.00 which were bigger than three (>3), suggesting that fish just arrived freshwater environment not long time then was soon caught.

The age of *C. nasus* at 2⁺ year was not much effect of the otolith size (the length of the bluish central) by the total length of each anchovy was (303-323 mm). The total length of fishes in this study were in range from 24cm - 36cm at 2 year that confirmed by Zhang et al. (2005) found that they were spawners in Yangtze River. A morphological and histological study on the ovary development of *C. nasus* demonstrated that the stages V and VI corresponded to the spawning and post-spawning periods (Li et al., 2007), meaning that the waters where both fish stages were captured should be the spawning areas (Jiang et al., 2017). Therefore, the *C. nasus* collected from water at Changshu section was move back to freshwater habitat to prepare for breeding and spawning with their gonad-matured reach at level II and III. So, *C. nasus* individuals were captured in the present study could be estimated that their spawning site will be far from the sampling site upstream along the lower reaches of the Yangtze River.

5. Conclusion

The otolith microchemistry analysis the value of Sr:Ca ratios of anchovy *C. nasus* collected from water around Changshu section was categorized into two types, in which five fishes had

different shorter and wider bluish central core regions. The shorter bluish core was in Type I while longer bluish core in Type II. All individuals are anadromous pattern of movement that utilized freshwater habitat in the early life then migrant to brackish water or seawater habitats of life history. One of them have an experience visiting in high salinity (marine water). After living in low salinity and next they move to other environment for foods and come back to freshwater habitat where they grow since was born to prepare spawning. According to the discussion in the point above in this chapter, the total body length and gonad maturity of each anchovy was clear to show that all samples were spawners and fish migrated to upper of the lower reaches of the Yangtze River for breeding and spawning. The habitat use and life history characteristic are different with the samples from Jiangjiang section and Chongming section, all of them utilized freshwater habitat in different degree in spawning ground and their whole life history.

General conclusion

The previous studies shown otolith microchemistry is an important for collecting the evidences of fish's life history events through the value of elements of otolith. At present, there have been many successful by using otolith elemental microchemistry to studies on migration history reconstruction of anadromous fish. The knowledge of migration patterns and habitat use strategies of the *C. nasus* species is very necessary for conservation and management in the future. In this study otolith microchemistry was used to determine habitat use strategies of *Coilia nasus* of the Jingjiang and Changshu sections in Jiangsu Province and Chongming section in Shanghai City in the lower reaches of the Yangtze River. The general findings were concluded that:

- *Coilia nasus* collected from the three sections in the lower reaches of the Yangtze River seemed to be different stocks and they spent early life period in freshwater habitat in different degree.
- *Coilia nasus* in the present study had flexible habitat use and life history types. Their early life spent in freshwater habitat, and then, they utilized freshwater, brackish water and marine water by complex life history strategies.
- The stock composition of the *C. nasus* was complex in the three sections of the Yangtze River. Almost all individuals were typical anadromous ones with different brackish and sea water experiences except for that two individuals were still with only freshwater experience yet.
- *Coilia nasus* collected from the three sections should be hatched from different natal origins. Those for the fish with high gonad maturity would be hatched near the estuary while those for the fish with low gonad maturity would be far from the estuary of the Yangtze River.

Innovations

1. Advanced otolith microchemistry technology reveals that *Coilia nasus* from the Jingjiang, Changshu and Chongming sections had flexible habitat use and life history characteristics.
2. The combined study of otolith microchemistry and biology of *C. nasus* in the present study clearly suggested that the resources of the fish were from different natal origins.
3. The Chongming section in estuary of the Yangtze River would be important up- and downstream migratory routes for both *C. nasus* with or without anadromous life history.

References

- [1] Arai T, Goto A, Miyazaki N. Migratory history of the threespine stickleback *Gasterosteus aculeatus*. Ichthyological Research. 2003. 50: 9–14
- [2] Arai T, Kotake A, Aoyama T, et al. Identifying sea-run brown trout, *Salmo trutta*, using Sr:Ca ratios of otolith. Ichthyological Research. 2002. 49: 380–383
- [3] Arrhenius F, and Hansson S. Growth and seasonal changes in energy content of young Baltic Sea herring (*Clupea harengus* L.). – ICES Journal of Marine Science. 1996. 53: 792–801.
- [4] Bath GE, Thorrold SR, Jones CM, et al. Strontium and barium uptake in aragonitic otoliths of marine fish. Geochimica et Cosmochimica Acta. 2000. 64, 1705–1714
- [5] Beamish R J, Chilton D E. Age determination methods for fishes studied by the Groundfish Program at the Pacific Biological Station. Canadian Special Publication of Fisheries and Aquatic Sciences. 1982. 60: 102 p.
- [6] Begg GA, Campana SE, Fowler AJ, et al. Otolith research and application: current directions in innovation and implementation. Marine and Freshwater Research. 2005. 56:477–483
- [7] Begg GA, Friedland KD, Pearce JB. Stock identification and its role in stock assessment and fisheries management: an overview. Fisheries Research. 1999. 43: 1–8.
- [8] Benson LK, Fitzsimons JM. Life history of the Hawaiian fish *Kuhlia sand vicensis* as inferred from daily growth rings of otoliths. Environmental Biology of Fishes. 2002. 65, 131–137.
- [9] Beverton RJH, Holt SJ. On the dynamics of exploited fish populations. Fisheries Investigations 1957, 19:1–533
- [10] Bolle L J, Rijnsdorp A D, Van Wim N, et al. Growth changes in plaice, cod, haddock and saithe in the North Sea: a comparison of (post-)medieval and present-day growth rates based on otolith measurements. Journal of Sea Research. 2004. 51: 313–328.
- [11] Pracheil B M, Hogan J D, Lyons J, et al. McIntyre. Using Hard-Part Microchemistry to Advance Conservation and Management of North American Freshwater Fishes, Fisheries. 2014. 39:10, 451–465.
- [12] Green B S, Mapstone B D, Carlos G., et al. Tropical Fish Otoliths: Information for Assessment, Management and Ecology. Reviews: Methods and Technologies in Fish Biology and Fisheries, Volume 11. 2009. DOI 10.1007/978-1-4020-5775-5
- [13] Brown R J and Severin K P. Otolith chemistry analyses indicate that water Sr:Ca is the primary factor influencing otolith Sr:Ca for freshwater and diadromous fish but not for marine fish. Canadian Journal of Fisheries and Aquatic Sciences. 2009. 66(10):1790–1808
- [14] Cadrin S X, Kerr L A, Mariani S (eds.). Stock identification methods: applications in fishery science. Elsevier Academic Press, London. 2014. 2nd ed., 566 p
- [15] Campana S E. Chemistry and composition of fish otoliths: pathways, mechanisms and applications. Marine Ecology Progress Series. 1999. 188: 263–297.
- [16] Campana S E, Chouinard G A, Hanson M, et al. Otolith elemental fingerprints as biological tracers of fish stocks. Fisheries Research. 2000. 46, 343e357.
- [17] Campana S E. Otolith science entering the 21st century. Marine and Freshwater Research. 2005. 56:485–495.
- [18] Campana S E, Neilson J D. Microstructure of fish otoliths. Canadian Journal of Fisheries and Aquatic Sciences. 1985. 42:1014–1032

- [19] Campana S E, Fowler A J, Jones C M. Otolith elemental fingerprinting for stock identification of Atlantic cod (*Gadus morhua*) using laser ablation ICPMS. Canadian Journal of Fisheries and Aquatic Sciences. 1994. 51, 1942–1950.
- [20] Campana S E, Thorrold S R. Otoliths, increments, and elements: keys to a comprehensive understanding of fish populations. Canadian Journal of Fisheries and Aquatic Sciences. 2001. 58:30–38
- [21] Campana S E. Otolith elemental composition as a natural marker of fish stocks. In: Cadrin, S.X., et al. (Eds.), Stock Identification Methods. Academic Press, New York, 2005. 227–245.
- [22] Carlson A K, Phelps Q E, Graeb B D S. Chemistry to conservation: using otoliths to advance recreational and commercial fisheries management. Journal of Fisheries Biology. 2016. 90(2): 505–527.
- [23] Chen TT, Jiang T, Liu HB, et al. Do all long supermaxilla-type estuarine tapertail anchovies (*Coilia nasus* Temminck et Schlegel, 1846) migrate anadromously? Journal of Applied Ichthyology. 2017. 33: 270–273. <https://doi.org/10.1111/jai.13309>
- [24] Chen T T., Jiang T, Li M M, et al. Inversion of habitat history for the long-jaw ecotype *Coilia nasus* collected from Nanjing section of the Yangtze River. Journal of Fisheries of China. 2016. 40, 882–892. (In Chinese with English abstract).
- [25] Chen Y Y and Chang J B. Wetlands loss and alteration of the floodplain in the lower and middle reaches of Yangtze River. In: The study of wetland in China. Y. Y. Chen (Ed.). Science and Technology Press, Jilin. 1995. 153–160.
- [26] Chen Y, Gong Q, Huang W, et al. A study on the characteristics and status of fishery resource in the Yangtze estuary. Journal of Fishery Sciences of China. 1999. 6, 48–51 (in Chinese with English abstract)
- [27] Chen X, Zong Y, Zhang E, et al. Human impacts on the Changjiang (Yangtze) River basin, China, with special reference to the impacts on the dry season water discharges into the sea. Geomorphology. 2001. 41 (2), 111–123.
- [28] Chen Z, Li J, Shen H, Wang Z. Yangtze River of China: historical analysis of discharge variability and sediment flux. Geomorphology. 2001. 41 (2), 77–91.
- [29] Secor DH. Specifying divergent migrations in the concept of stock: the contingent hypothesis. Fisheries Research, Volume 43, Issues 1–3, 1999, Pages 13–34, [https://doi.org/10.1016/S0165-7836\(99\)00064-8](https://doi.org/10.1016/S0165-7836(99)00064-8).
- [30] Dominique Alò, Cristian Correa, Horacio Samaniego, et al. Otolith microchemistry and diadromy in Patagonian river fishes. PeerJ. 2019. 7: e6149. doi: 10.7717/peerj.6149
- [31] Dou S, Kazuki Y, Yu X, et al. The migratory history of anadromous and non-anadromous tapertail anchovy *Coilia nasus* in the Yangtze River estuary revealed by the otolith Sr:Ca ratio. Environmental Biology of Fishes. 2012. 95, 481–490.
- [32] Elsdon T S, Wells B K, Campana S E, et al. Otolith chemistry to describe movements and life-history parameters of fishes: hypotheses, assumptions, limitations and inferences. Journal of Experimental Marine Biology name. 2008. 1:297–330.
- [33] FAO. The State of World Fisheries and Aquaculture 2018 - Meeting the sustainable development goals. Rome. 2018. Licence: CC BY-NC-SA 3.0 IGO.
- [34] Farrell J, Campana SE. Regulation of calcium and strontium deposition on the otoliths of juvenile tilapia, *Oreochromis niloticus*. Comparative Biochemistry and Physiology A-Molecular & Integrative Physiology. 1996. 115A, 103–109.
- [35] Fey D P, Jan W Weslawski. Age, growth rate, and otolith growth of polar cod (*Boreogadus saida*) in two fjords of Svalbard, Kongsfjorden and Rijpfjorden. Oceanology. 2017. 59(4):576–584.

- [36] Fowler A J. Age in Years from Otoliths of Adult Tropical Fish. In: Green BS, Mapstone BD, Carlos G, Begg GA (eds). Tropical Fish Otoliths: Information for Assessment, Management and Ecology. Reviews: Methods and Technologies in Fish Biology and Fisheries, Springer. vol 11. 2009. 55–92
- [37] Fowler A J, Campana SE, Jones CM, et al. Experimental assessment of the effect of temperature and salinity on elemental composition of otoliths using solution-based ICPMS. Canadian Journal of Fisheries and Aquatic Sciences. 1995. 52, 1421–1430.
- [38] Fu C Z, Wu J H, Chen J K, et al. Freshwater fish biodiversity in the Yangtze River basin of China: Patterns, threats and conservation. Biodiversity & Conservation. 2003. 12, 1649–1685.
- [39] Fu C Z, Wu J H, Wang X Y, et al. Patterns of diversity, altitudinal range and body size among freshwater fishes in the Yangtze River basin, China. Global Ecology and Biogeography. 2004. 13, 543–552.
- [40] Florida Fish and Wildlife Conservation Commission (2021). Introduction to Ageing Fish: What Are Otoliths? <https://myfwc.com/research/saltwater/fish/age-growth-lab/ageing-fish-otoliths/>
- [41] Geffen AJ, Pearce NJG, Perkins WT. Metal concentrations in fish otoliths in relation to body composition after laboratory exposure to mercury and lead. Marine Ecology Progress Series. 1998. 165, 235–245.
- [42] Gillanders B M. Otolith chemistry to determine movements of diadromous and freshwater fish. Aquatic Living Resources. 2005. 18:291-300.
- [43] Green BS, Mapstone BD, Carlos G, et al. Introduction to Otoliths and Fisheries in the Tropics. In: Green BS, Mapstone BD, Carlos G, Begg GA (eds). Tropical Fish Otoliths: Information for Assessment, Management and Ecology. Reviews: Methods and Technologies in Fish Biology and Fisheries, Springer. vol 11. 2009.
- [44] Gao Y W, Joner S H, Bargmann G G. Stable isotopic composition of otoliths in identification of spawning stocks of Pacific herring (*Clupea pallasii*) in Puget Sound. Canadian Journal of Fisheries and Aquatic Sciences. 2001. 58: 2113–2120.
- [45] Hobbs J A, Yin Q, Burton J. et al. Retrospective determination of natal habitats for an estuarine fish with otolith strontium isotope ratios. Marine and Freshwater Research. 2005. 56: 655–660.
- [46] Jiang T, Liu H, Lu M, et al. A possible connectivity among estuarine tapertail anchovy (*Coilia nasus*) populations in the Yangtze River, Yellow Sea, and Poyang Lake. Estuaries and Coasts. 2016. 39(6):1762–1768.
- [47] Jiang T, Yang J, Liu H, et al. Life history of *Coilia nasus* from the Yellow Sea inferred from otolith Sr:Ca ratios. Environmental Biology of Fishes. 2012. 95(4):503–508.
- [48] Jiang T, Yang J, Lu MJ, et al. Discovery of a spawning area for anadromous *Coilia nasus* Temminck et Schlegel, 1846 in Poyang Lake, China. Journal of Applied Ichthyology. 2017. 33:189–192.
- [49] Jiang Tao, Liu Hongbo, Shen Xinqiang, et al. Life history variations among different populations of *Coilia nasus* along the Chinese coast inferred from Otolith microchemistry. Journal of the Faculty of Agriculture, Kyushu University. 2014. 59(2): 383–389
- [50] Jones C M. Development and application of the otolith increment technique. In: D K Stevenson and S E Campana (eds.), Otolith microstructure examination and analysis. In: Canadian Special Publication of Fisheries and Aquatic Sciences. 1992. 117, 1–11
- [51] Kennedy B D, Folt CL, Blum JD, et al. Natural isotope markers in salmon. Nature. 1997. 387, 766–767.
- [52] Kennedy B P, Blum J D, Folt C L, et al. Using natural strontium isotopic signatures as fish markers: methodology and application. Canadian Journal of Fisheries and Aquatic Sciences. 2000. 57, 2280–2292

- [53] Kennedy B P, Klaue A, Blum J D, et al. Reconstructing the lives of fish using Sr isotopes in otoliths. *Canadian Journal of Fisheries and Aquatic Sciences*. 2002. 59, 925–929.
- [54] Kerr L A, Campana S E. Chemical composition of fish hard parts as a natural marker of fish stocks. Elsevier Academic Press, 2014. 205–234.
- [55] Kerr LA, Secor DH, Kraus RT. Stable isotope (d13C and d18O) and Sr/Ca composition of otoliths as proxies for environmental salinity experienced by an estuarine fish. *Marine Ecology Progress Series*. 2007. 349, 245–253.
- [56] Khumbanyiwa D D, Mengmeng L, Tao J, et al. Unraveling habitat use of *Coilia nasus* from Qiantang River of China by otolith microchemistry. *Regional Studies in Marine Science* 18 (2018) 122–128.
- [57] Li K. Management and restoration of fish communities in Lake Taihu, China. *Fisheries Management and Ecology*. 1999. 6, 71–81.
- [58] Li Y X, Xie S G, Li Z J, et al. Gonad development of an anadromous fish *Coilia ectenes* (Engraulidae) in lower reach of Yangtze River, China. *Fisheries Science*. 2007. 73, 1224–1230.
- [59] Limburg K E, Olson C, Walther Y, et al. Tracking Baltic hypoxia and cod migration over millennia with natural tags. *Proceedings of the National Academy of Sciences of the United States of America*. 2011. 108: E177–E182
- [60] Lisa A Kerr, Campana Steven E. Chapter Eleven - Chemical Composition of Fish Hard Parts as a Natural Marker of Fish Stocks, Editor(s): Steven X Cadrin, Lisa A Kerr, Stefano Mariani, *Stock Identification Methods (Second Edition)*, Academic Press. 2014. 205–234.
- [61] Liu H, Jiang T, Tan X, et al. Preliminary investigation on otolith microchemistry of naked carp (*Gymnocypris przewalskii*) in Lake Qinghai, China. *Environmental Biology of Fishes*. 2012. 95(4):455–461.
- [62] Liu Hongbo, Jiang Tao, Yang Jian. Unravelling habitat use of *Coilia nasus* from the Rokkaku River and Chikugo River estuaries of Japan by otolith strontium and calcium. *Acta Oceanologica Sinica*. 2018. 37(6): 52–60.
- [63] Liu Kai, Duan Jinrong, Xu Dongpo, et al. Present situation of *Coilia nasus* population features and yield in Yangtze River estuary waters in fishing season. *Chinese Journal of Ecology*. 2012. 31(12): 3138 – 3143.
- [64] Liu K, Zhang M, Xu D, et al. Studies on resource change and MSY of *Coilia mystus* in the Yangtze River estuary. *Journal of Shanghai Fisheries University*. 2004. 13, 298–303.
- [65] Liu Dong, Li Yingying, Tang Wenqiao, et al. Population structure of *Coilia nasus* in the Yangtze River revealed by insertion of short interspersed elements. *Biochemical Systematics and Ecology* 54 (2014) 103–112.
- [66] Lochet A, Limburg KE, Rudstam L, et al. Selenium incorporation in fish otoliths: effects of selenium and mercury from the water. *Canadian Journal of Fisheries and Aquatic Sciences*. 2010. 67, 1388–1397.
- [67] Ma C, Liu M, Ma L, et al. Genetic diversity in *Coilia ectenes* by RAPD analysis. *Marine Fisheries Research*. 2004. 25, 19–24.
- [68] Milton DA, Chenery SR. Sources and uptake of trace metals in otoliths of juvenile barramundi (*Lates calcarifer*). *Journal of Experimental Marine Biology and Ecology*. 2001. 264, 47–65.
- [69] Morales-Nin B. (2000) Review of the growth regulation processes of otolith daily increment formation. *Fisheries Research*. 2000. 46:53–67
- [70] Morris Jr J A, Rulifson R A, Toburen L H. Life history strategies of striped bass, *Morone saxatilis*, and populations inferred from otolith microchemistry. *Fisheries Research*. 2003. 62: 53–63.

- [71] Pavlovich M A, Greer C E. Yangtze River. Encyclopedia Britannica. Encyclopedia Britannica, inc. December 29, 2019. <https://www.britannica.com/place/Yangtze-River>.
- [72] Papetti C, Franco Di, Zane A, et al. Single population and common natal origin for Adriatic Scomber scombrus stocks: evidence from an integrated approach. ICES Journal of Marine Science. 2013. 70, 387–398.
- [73] Popper A N, Lu Z. Structure-function relationships in fish otolith organs. Fisheries Research. 2000. 46(1-3):15-25
- [74] Popper A N, Ramcharitar J, Campana SE. Why otoliths? Insights from inner ear physiology and fisheries biology. Marine and Freshwater Research. 2005. 56:97–504
- [75] Pracheil BM, Hogan JD, Lyons J, et al. Using hard-part microchemistry to advance conservation and management of North American freshwater fishes. Fisheries. 2014. 39:451–465.
- [76] Rodionov SN. A sequential algorithm for testing climate regime shifts. Geophysical Research Letters. 2004. 31, doi: 10.1029/2004GL019448.
- [77] Rodionov SN. Regime Shift Detection Calculator Now Available on the Bering Climate. 2005. Website. <https://www.arcus.org/arctic-info/archive/17132>
- [78] Ru H J, Liu X Q. River-lake migration of fishes in the Dongting Lake area of the Yangtze floodplain. Geophysical Research Letters. 29 (2013), 594–601.
- [79] Secor D H, Rooker J R. Is otolith strontium a useful scalar of life–cycles in estuarine fishes? Fisheries Research. 2000. 46: 359–371.
- [80] Secor DH, Rooker JR, Zlokovitz E, et al. Identification of riverine, estuarine, and coastal contingents of Hudson River striped bass based upon otolith elemental fingerprints. Marine Ecology Progress. 2001. 211, 245–253.
- [81] Shi X, Qin B. Evolution and ecological environment of lakes in the middle and lower reaches of Yangtze River. Journal of Ningbo University. 2007. 20, 221–226.
- [82] Sokta Ly, Tao Jiang, Hongbo Liu, et al. Loss of *Coilia nasus* habitats in Chinese freshwater lakes: An otolith microchemistry assessment. Heliyon 6 (2020) e04571.
- [83] Spencer K, Shafer DJ, Gauldie RW, et al. Stable lead isotope ratios from distinct anthropogenic sources in fish otoliths: a potential nursery ground stock marker. Comparative Biochemistry and Physiology A-Molecular & Integrative Physiology. 2000. 127A, 273–284.
- [84] Sponaugle S. Daily otolith increments in the early stages of tropical fishes. In: B Green, B Mapstone, G Carlos, and G Begg, eds. Gathering information from otoliths of tropical fishes. Methods and Technologies in Fish Biology and Fisheries. Springer. 2009. 93-132
- [85] Sponaugle S. Otolith microstructure reveals ecological and oceanographic processes important to fisheries management. Fishes Environmental Biology of Fishes. 2010. 89: 221-238
- [86] Sturrock M, Trueman C N, Darnaude A M, et al. Can otolith elemental chemistry retrospectively track migrations in fully marine fishes? Journal of Fish Biology (2012) 81, 766–795.
- [87] Thorrold SR, Swearer SE. Otolith chemistry. In: Green BS, Mapstone BD, Carlos G, Begg GA (eds). Tropical fish otoliths: information for assessment, management and ecology. Springer. 2009. 249–295
- [88] Thorrold SR, Latkoczy C, Swart PK, et al. Natal homing in a marine fish metapopulation. Science. 2001. 291, 297–299.
- [89] Thorrold SR, Jones CM, Campana SE, et al. Trace element signatures in otoliths record natal river of juvenile American shad (*Alosa sapidissima*). Limnology and Oceanography. 1998a. 43, 1826–1835.

- [90] Thorrold SR, Jones CM, Swart PK, et al. Accurate classification of juvenile weakfish *Cynoscion regalis* to estuarine nursery areas based on chemical signatures in otoliths. *Marine Ecology Progress Series*. 1998b. 173, 253e265
- [91] Thresher RE. Elemental composition of otoliths as a stock delineator in fishes. *Fisheries Research*. 1999. 43, 165–204.
- [92] Tsukamoto K, Chow S, Otake T, et al. Oceanic spawning ecology of freshwater eels in the western North Pacific. *Nature Communications*. 2011. 2:179
- [93] Wang Q, Zhou L, Little SH, et al. The geochemical behavior of Cu and its isotopes in the Yangtze River. *Science of the Total Environment* 2020, 728:138428.
- [94] Wang M, Xu G, Tang Y, et al. Transcriptome analysis of the brain provides insights into the regulatory mechanism for *Coilia nasus* migration. *BMC Genomics* 21, 410 (2020).
- [95] Wang Meiyao, Xu Pao, Zhu Zhixiang. Regulation of signal transduction in *Coilia nasus* during migration, *Genomics*, 112. 2020. 112(1):55-64.
- [96] Weber PK, Hutcheon ID, McKeegan KD, et al. Otolith sulfur isotope method to reconstruct salmon (*Oncorhynchus tshawytscha*) life history. *Canadian Journal of Fisheries and Aquatic Sciences* 2002. 59:587–591.
- [97] Whitehead P J P, Nelson G J, Wongratana T. *FAO Species Catalogue. Vol. 7. Clupeoid Fishes of the World (Suborder lupeoidei): an annotated and illustrated catalogue of the herrings, sardines, pilchards, sprats, shads, anchovies, and wolfherrings: Part 2. Engraulidae*. Rome: *FAO Fisheries Synopsis No.* 1988. 125, 460–475
- [98] Wilcove D S, Wikelski M. Going, going, gone: is animal migration disappearing. *Plos Biology* 6, 2008, e188.
- [99] Wirgin I, Breece M W, Fox D A, et al. Origin of Atlantic sturgeon collected off the Delaware coast during spring months. *North American Journal of Fisheries Management*. 2015. 35, 20– 30.
- [100] Xu G., Bian C., Nie Z., et al. Genome and population sequencing of a chromosome-level genome assembly of the Chinese tapertail anchovy (*Coilia nasus*) provides novel insights into migratory adaptation. *GigaScience*, 2020, 9:1–13
- [101] Yang J, Arai T, Liu H, et al. Reconstructing habitat use of *Coilia mystus* and *Coilia ectenes* of the Yangtze River Estuary, and of *Coilia ectenes* of Taihu Lake, based on otolith strontium and calcium. *Journal of Fish Biology*. 2006. 69: 1120–1135.
- [102] Yang J, Jiang T, Lu M, et al. First identification of a possible nursery area for diadromous *Coilia nasus* in the Poyang Lake nearly 1000km away from the Yangtze River Estuary. *Front. Mar. Sci. Conference Abstract: XV European Congress of Ichthyology*. 2015. doi: 10.3389/conf.fmars .2015.03.00040
- [103] Yang G S, Ma C D, & Chang S Y. *Yangtze conservation and development report 2009*. Wuhan: Changjiang Press. 2009.
- [104] Yang G S, Wen L D, & Li L F. *Yangtze conservation and development report 2007*. Wuhan: Changjiang Press. 2007.
- [105] Yang G S, Zhu C Q, & Jiang Z G. *Yangtze conservation and development report 2011*. Wuhan: Changjiang Press. 2011.
- [106] Yang J, Jiang T, Liu H. Are there habitat salinity markers of the Sr:Ca ratio in the otolith of wild diadromous fishes? A literature survey. *Ichthyological Research* (2011) 58:291–294.
- [107] Ye S W, Li Z J, Liu J S, et al. Distribution, endemism and conservation status of fishes in the Yangtze River basin, China. In: Grillo O., editor. *Ecosystems Biodiversity*. InTech; Rijeka, Croatia. 2011. 41–66

- [108] Yuan C, Lin J, Qin A, et al. On the classification history and status quo of genus *Coilia* in China. J Nanjing University. 1976. 2:1–12.
- [109] Yuan C, Qin A. Ecological habits and distribution of *Coilia* along the Chinese coast and its changes of output. Marine Science. 1984. 5, 35–37.
- [110] Yuan C. Spawning migration of *Coilia nasus*. Bulletin of Biology. 1987.12: 1–3.
- [111] Zeigler J M, Whitledge G. Assessment of otolith chemistry for identifying source environment of fishes in the Lower Illinois River, Illinois. Hydrobiologia 638, No. 1. 2010. 109-119.
- [112] Zhang H, Kang M, Shen L, et al. Rapid change in Yangtze fisheries and its implications for global freshwater ecosystem management. Fish and Fisheries. 2020. 00:1–20.
- [113] Zhang MY, Xu DP, Liu K, et al. Studies on biological characteristics and change of resource of *Coilia nasus* Schlegel in the lower reaches of the Yangtze River (in Chinese), Resources and Environment in the Yangtze Basin. 14 (2005) 694–698,
- [114] Zhang SY (Ed.). Fauna Sinica-osteichthyes: Acipenseriformes, Elopiformes, Clupeiformes, Gonorhynchiformes. Science Press, Beijing. 2001. 148–156.
- [115] Zheng Y, Guo H, Wei K, et al. Micro-PIXE study of whole otolith of *Anguilla japonica* at elver stage. Nuclear Instruments and Methods in Physics Research B. 2010. 268: 2152–2155.

APPENDIX

Experimental design

